

## Decision Tree

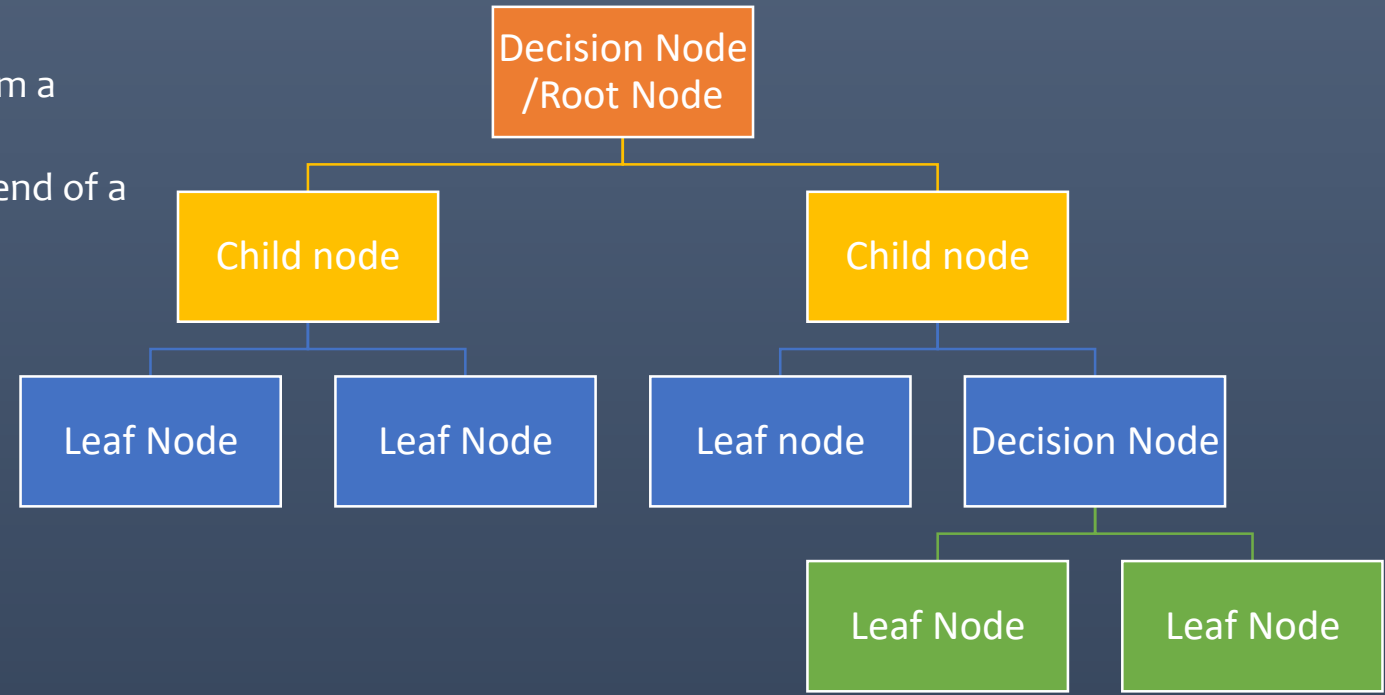
This is a tree-based approach and A Decision Tree is a Supervised learning technique that can be used for classification and Regression problems,

however, it is mostly preferred for solving Classification problems. It is a tree-structured classifier, where internal nodes represent the features of a dataset, branches represent the decision rules and each leaf node represents the outcome

**Root nodes:** The top-level nodes in a tree structure from which all other nodes descend.

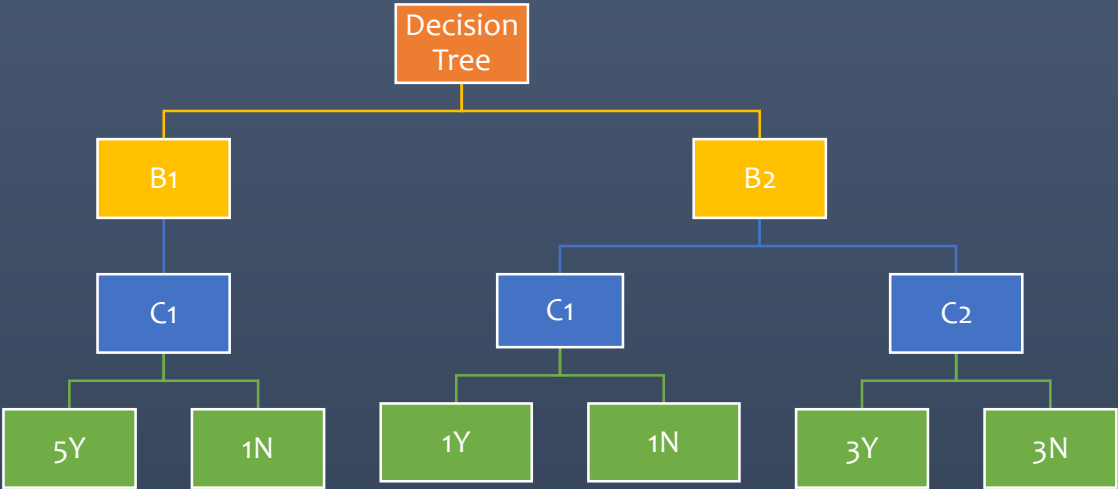
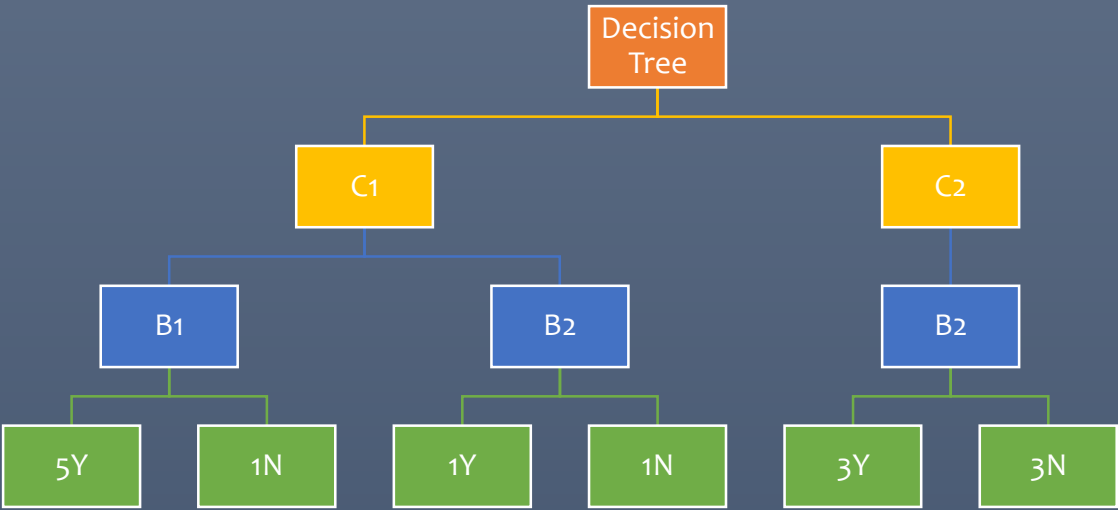
**Child nodes:** Nodes that are directly connected and descend from a parent node.

**Leaf nodes:** Nodes that have no children and are located at the end of a tree branch.



| Feature 01 | Feature 02 | Yes/No |
|------------|------------|--------|
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B2         | Y      |
| C2         | B2         | Y      |
| C2         | B2         | Y      |
| C2         | B2         | Y      |
| C1         | B1         | N      |
| C1         | B2         | N      |
| C2         | B2         | N      |
| C2         | B2         | N      |
| C2         | B2         | N      |

When constructing the tree, the model must choose one feature among multiple features as the parent node. This selection is based on the concept of **information gain**, where the feature with the highest information gain becomes the parent node, while the remaining features become the child nodes.



It is basically **splitting the data into a tree-like structure**.

A Decision Tree has three main parts:

- **Root Node** – the starting point of the tree.
- **Child Nodes** – the branches created after splitting the data.
- **Leaf Nodes** – the final result or prediction.

Since there are multiple features in the dataset, we need to decide **which feature should be the Root Node** and which features should come next as Child Nodes.

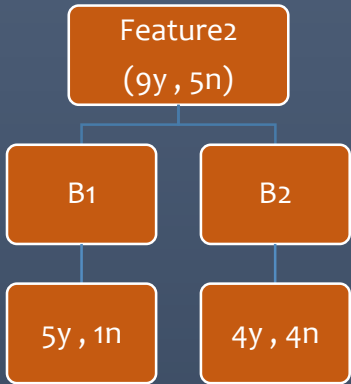
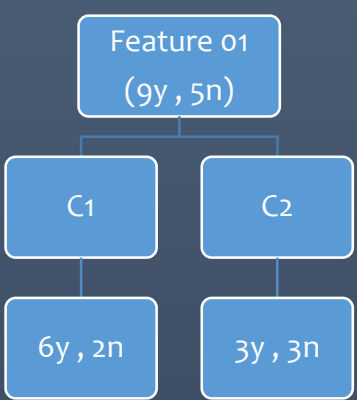
The feature with the **highest Information Gain** is selected as the **Root Node**. The other features are then used for the subsequent child nodes.

#### **Steps to Calculate Information Gain**

1. First, calculate **Entropy** or **Gini Impurity**.
2. Use the calculated value in the **Information Gain formula**.
3. Calculate the Information Gain for **each feature**.
4. Compare the Information Gain of all features.
5. The feature with the **highest Information Gain becomes the Root Node**.
6. The remaining features are used to create the **subsequent Child Nodes**.

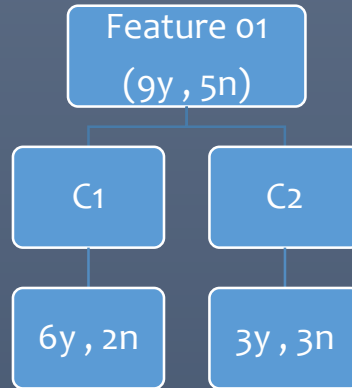
| Feature 01 | Yes/No |
|------------|--------|
| C1         | Y      |
| C1         | Y      |
| C1         | Y      |
| C1         | Y      |
| C1         | Y      |
| C1         | Y      |
| C2         | Y      |
| C2         | Y      |
| C2         | Y      |
| C1         | N      |
| C1         | N      |
| C2         | N      |
| C2         | N      |
| C2         | N      |

| Feature 02 | Yes/No |
|------------|--------|
| B1         | Y      |
| B1         | Y      |
| B1         | Y      |
| B1         | Y      |
| B1         | Y      |
| B2         | Y      |
| B2         | Y      |
| B2         | Y      |
| B2         | Y      |
| B1         | N      |
| B2         | N      |
| B2         | N      |
| B2         | N      |
| B2         | N      |



## Calculate the Entropy of Feature One

$$Entropy = - \sum_{i=1}^N p_i \log_2 p_i$$



### Entropy of Root Node(9y , 5n)

$$\begin{aligned}
 Entropy &= (-P_y) * (\log_2(P_y)) + (-P_n) * (\log_2(P_n)) \\
 &= \left(-\left(\frac{9}{14}\right)\right) * \left(\log_2\left(\frac{9}{14}\right)\right) + \left(-\left(\frac{5}{14}\right)\right) * \left(\log_2\left(\frac{5}{14}\right)\right) \\
 &= (-0.64) * (\log_2(0.64)) + (-0.35) * (\log_2(0.35)) \\
 &= (-0.64) * (-0.64) + (-0.35) * (-1.49) \\
 \textbf{Entropy} &= \textbf{0.94}
 \end{aligned}$$

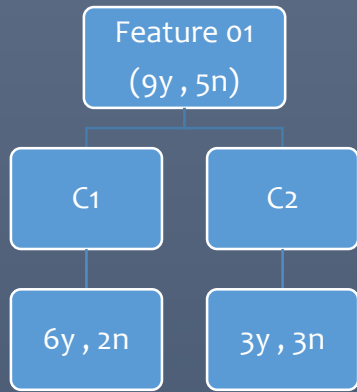
### Entropy of Left Node (6y , 2n)

$$\begin{aligned}
 Entropy &= -P_y \log_2(P_y) - P_n \log_2(P_n) \\
 &= \left(-\left(\frac{6}{8}\right)\right) * \left(\log_2\left(\frac{6}{8}\right)\right) + \left(-\left(\frac{2}{8}\right)\right) * \left(\log_2\left(\frac{2}{8}\right)\right) \\
 &\dots\dots\dots \\
 \textbf{Entropy} &= \textbf{0.81}
 \end{aligned}$$

### Entropy of a Right Node(3y , 3n)

$$\begin{aligned}
 Entropy &= -P_y \log_2(P_y) - P_n \log_2(P_n) \\
 &= \left(-\left(\frac{3}{6}\right)\right) * \left(\log_2\left(\frac{3}{6}\right)\right) + \left(-\left(\frac{3}{6}\right)\right) * \left(\log_2\left(\frac{3}{6}\right)\right) \\
 &\dots\dots\dots \\
 \textbf{Entropy} &= \textbf{1}
 \end{aligned}$$

## Gini Impurity the Entropy of Feature One



$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

**GI of Root Node(9y , 5n)**

$$\begin{aligned} Gini &= 1 - \left[ \left( \frac{9}{14} \right)^2 + \left( \frac{5}{14} \right)^2 \right] \\ &= 1 - [ (0.64)^2 + (0.36)^2 ] \\ &= 1 - [ 0.41 + 0.13 ] \\ \mathbf{Gini} &= \mathbf{0.46} \end{aligned}$$

**GI of Left Node (6y , 2n)**

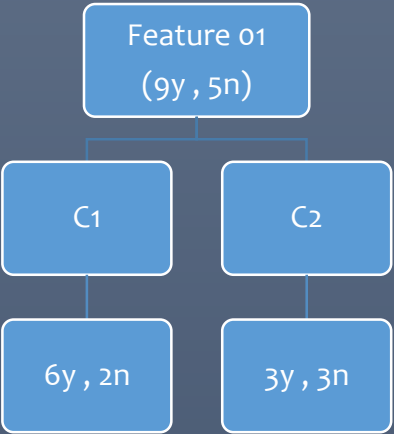
$$\begin{aligned} Gini &= 1 - \left[ \left( \frac{6}{8} \right)^2 + \left( \frac{2}{8} \right)^2 \right] \\ &= 1 - [ (0.75)^2 + (0.25)^2 ] \\ &= 1 - [ 0.56 + 0.06 ] \\ \mathbf{Gini} &= \mathbf{0.38} \end{aligned}$$

**GI of Right Node(3y , 3n)**

$$\begin{aligned} Gini &= 1 - \left[ \left( \frac{3}{6} \right)^2 + \left( \frac{3}{6} \right)^2 \right] \\ &= 1 - [ (0.50)^2 + (0.50)^2 ] \\ &= 1 - [ 0.25 + 0.25 ] \\ \mathbf{Gini} &= \mathbf{0.50} \end{aligned}$$

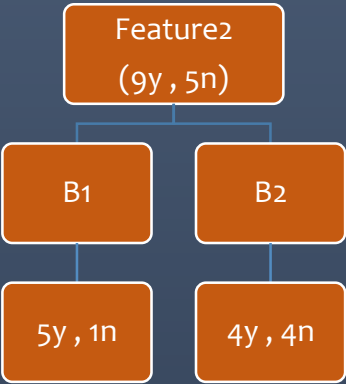
Entropy and Gini Impurity the Entropy of Feature One

|               | Root Node | Child Node | Child Node |
|---------------|-----------|------------|------------|
| Feature One   | (9y , 5n) | (6y , 2n)  | (3y , 3n)  |
| Entropy       | 0.94      | 0.81       | 1          |
| Gini Impurity | 0.46      | 0.38       | 0.5        |



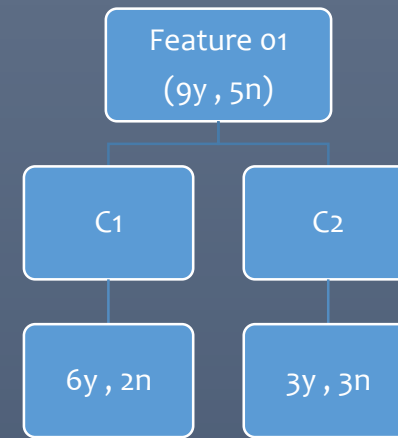
Entropy and Gini Impurity the Entropy of Feature Two

|               | Root Node | Child Node | Child Node |
|---------------|-----------|------------|------------|
| Feature Two   | (9y , 5n) | (5y , 1n)  | (4y , 4n)  |
| Entropy       | 0.94      | 0.65       | 1          |
| Gini Impurity | 0.48      | 0.28       | 0.5        |



## Calculate the Information Gain of the Feature One

|               | Root Node | Child Node | Child Node |
|---------------|-----------|------------|------------|
| Feature One   | (9y , 5n) | (6y , 2n)  | (3y , 3n)  |
| Entropy       | 0.94      | 0.81       | 1          |
| Gini Impurity | 0.46      | 0.38       | 0.5        |



$$\text{Information Gain}(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

$H(S)$  = Entropy or Gini of the root Node.  
 $Sv$  = total values in a particular node.  
 $S$  = Total Number of Values of Output Feature  
 $H(Sv)$  = Entropy or Gini of the Child Node

### IG Using Entropy

$$\text{Gain}(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

$$\text{Gain}(S, f1) = H(S) - \left[ \left( \frac{|Sv|}{|S|} * H(Sv) \right) + \left( \frac{|Sv|}{|S|} * H(Sv) \right) \right]$$

$$\text{Gain}(S, f1) = 0.94 - \left[ \left( \frac{8}{14} * 0.81 \right) + \left( \frac{6}{14} * 1 \right) \right]$$

$$\text{Gain}(S, f1) = 0.049$$

### IG Using GI

$$\text{Gain}(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

$$\text{Gain}(S, f1) = H(S) - \left[ \left( \frac{|Sv|}{|S|} * H(Sv) \right) + \left( \frac{|Sv|}{|S|} * H(Sv) \right) \right]$$

$$\text{Gain}(S, f1) = 0.46 - \left[ \left( \frac{8}{14} * 0.38 \right) + \left( \frac{6}{14} * 0.50 \right) \right]$$

$$\text{Gain}(S, f1) = 0.029$$

## Calculate the Information Gain of the Feature Two

|               | Root Node | Child Node | Child Node |
|---------------|-----------|------------|------------|
| Feature Two   | (9y , 5n) | (5y , 1n)  | (4y , 4n)  |
| Entropy       | 0.94      | 0.65       | 1          |
| Gini Impurity | 0.48      | 0.28       | 0.5        |

$$\text{Information Gain}(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

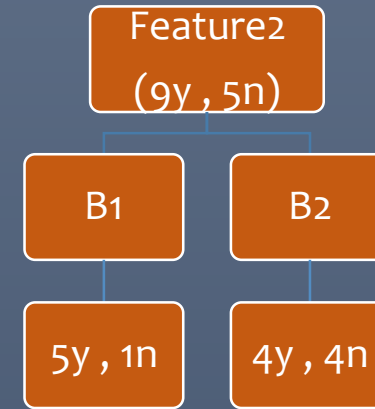
### IG Using Entropy

$$\text{Gain}(S, f2) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

$$\text{Gain}(S, f2) = H(S) - \left[ \left( \frac{|Sv|}{|S|} * H(Sv) \right) + \left( \frac{|Sv|}{|S|} * H(Sv) \right) \right]$$

$$\text{Gain}(S, f2) = 0.94 - \left[ \left( \frac{6}{14} * 0.65 \right) + \left( \frac{8}{14} * 1 \right) \right]$$

$$\text{Gain}(S, f2) = 0.090$$



$H(S)$  = Entropy or Gini of the root Node.

$Sv$  = total values in a particular node.

$S$  = Total Number of Values of Output Feature

$H(Sv)$  = Entropy or Gini of the Child Node

### IG Using GI

$$\text{Gain}(S, f1) = H(S) - \sum \frac{|Sv|}{|S|} H(Sv)$$

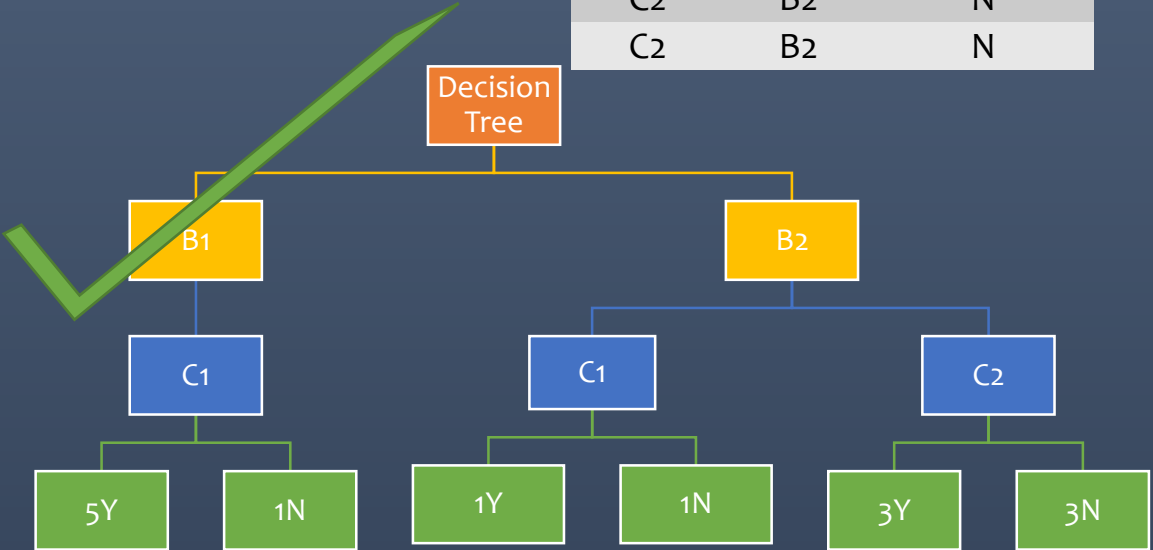
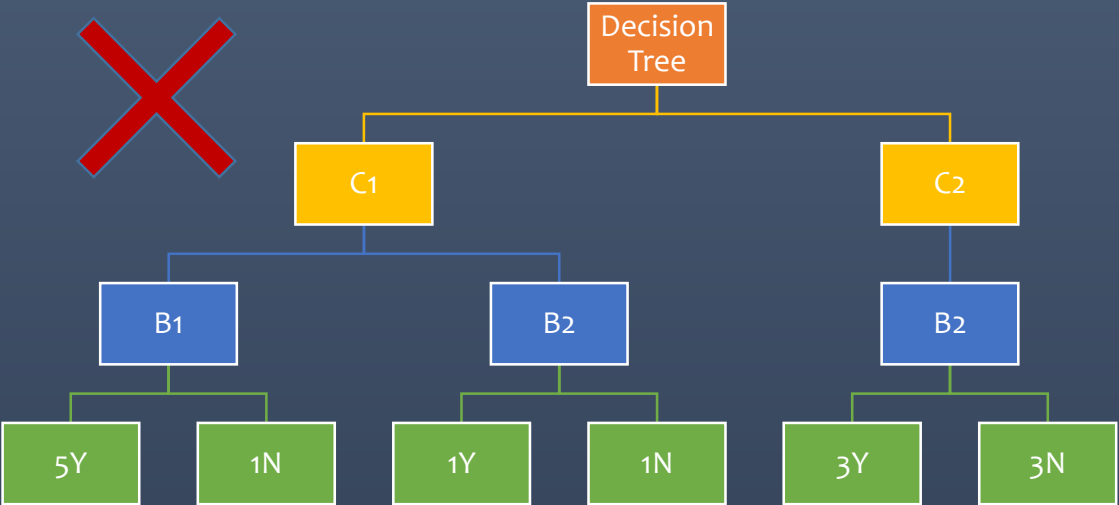
$$\text{Gain}(S, f1) = H(S) - \left[ \left( \frac{|Sv|}{|S|} * H(Sv) \right) + \left( \frac{|Sv|}{|S|} * H(Sv) \right) \right]$$

$$\text{Gain}(S, f1) = 0.46 - \left[ \left( \frac{6}{14} * 0.28 \right) + \left( \frac{8}{14} * 0.50 \right) \right]$$

$$\text{Gain}(S, f1) = 0.074$$

|             | Information Gain |               |
|-------------|------------------|---------------|
|             | Entropy          | Gini Impurity |
| Feature One | 0.049            | 0.029         |
| Feature Two | 0.090            | 0.074         |

| Feature 01 | Feature 02 | Yes/No |
|------------|------------|--------|
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B1         | Y      |
| C1         | B2         | Y      |
| C2         | B2         | Y      |
| C2         | B2         | Y      |
| C2         | B2         | Y      |
| C1         | B1         | N      |
| C1         | B2         | N      |
| C2         | B2         | N      |
| C2         | B2         | N      |
| C2         | B2         | N      |



Hyper parameter

**Criterion** Gini impurity or Entropy

max Features: Number of feature to be short listed for model training

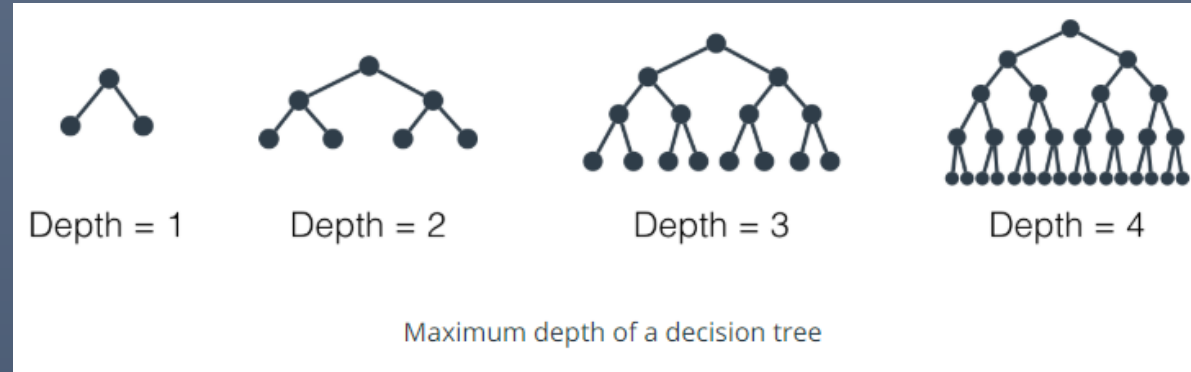
Example if you have 50 feature and for model training you keep only 10 feature as max feature you model will be trained on 10 features

## Hyper parameter

Max Depth

Min Sample Split:

Min Sample Leaf:



Hyper parameter tuning.

Max Depth:-

The height of a decision tree, also known as its depth, represents the number of levels or layers it contains.

Min Sample Split:

Minimum sample split decides minimum number of samples necessary to split a **nonterminal node**.

By default, the decision tree tries to split every node with two or more.

Min Sample Leaf:

it defines the minimum number of **data samples required to form a terminal leaf node** in a decision tree.

## Pruning

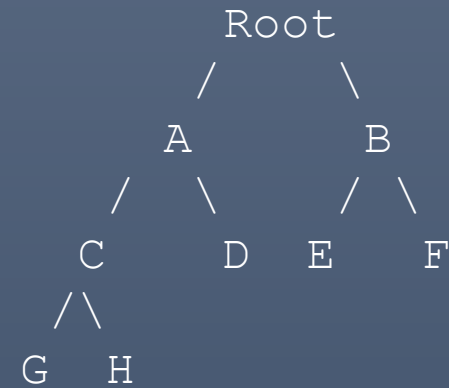
in a Decision Tree means **removing unnecessary branches/splits from a tree to reduce overfitting.**

The tree may become extremely complex and start **memorizing the training data** rather than learning general patterns.

Max Depth

Min Sample Split:

Min Sample Leaf:

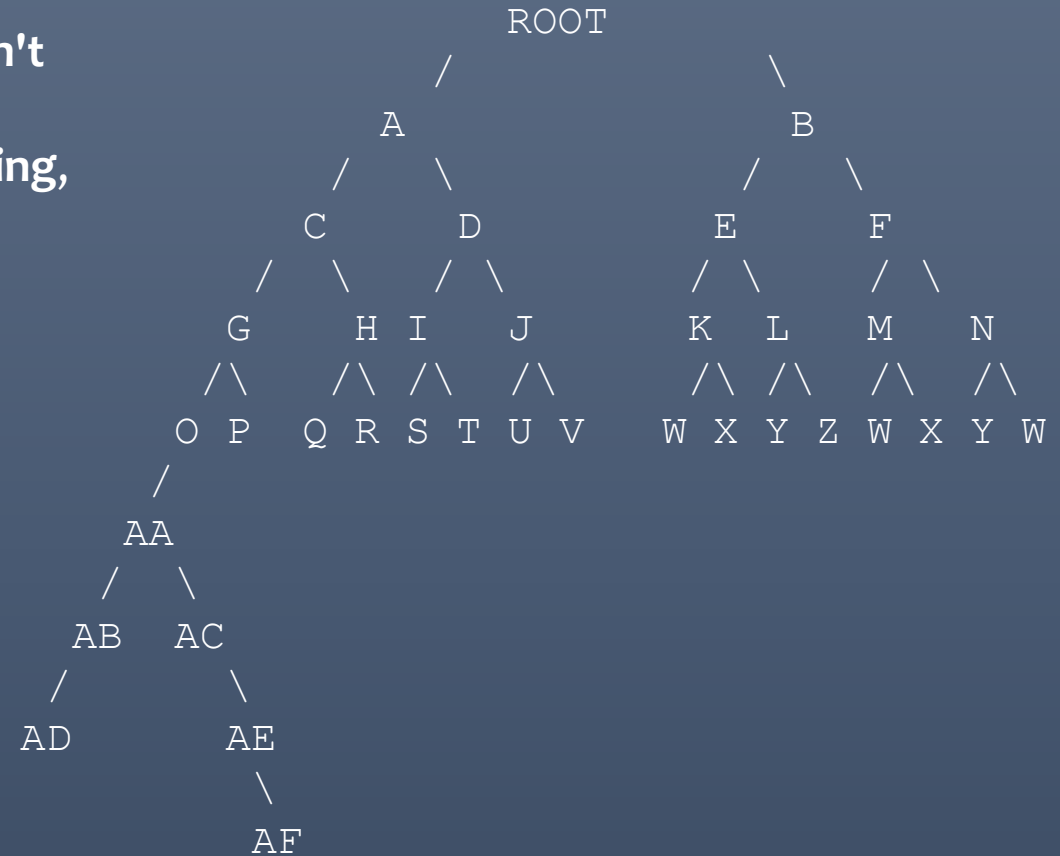


## Post Pruning:

First, you allow the tree to grow relatively large:

Then you examine the tree and **remove branches that aren't providing enough benefit.**

In scikit-learn, a common method is **Cost-Complexity Pruning**, controlled by: `ccp_alpha`



`DecisionTreeClassifier(ccp_alpha=0.0)`



Very large tree



Potential overfitting

`DecisionTreeClassifier(ccp_alpha=0.01)`



Some branches removed



May be best fit model

`DecisionTreeClassifier(ccp_alpha=0.1)`



Much simpler tree



Potential underfitting